

Tuple data type

A collection of objects enclosed within parentheses () separated by commas is called a Tuple. So we can say tuple is a sequential collection of elements.

1. Collection of objects enclosed within parentheses () is called a tuple.
2. Duplicates are allowed.
3. Ordered / Sequential collection.
4. Indexing supported.
5. Slicing supported.
6. Different data types allowed.
7. Immutable in nature (elements cannot be modified).

Example:-

```
t1 = (10, 20, 30, 40)
t2 = ("Python", "Java", "C++")
t3 = (10, "Python", 3.14, True)
```

```
print(t1)
print(t2)
print(t3)
```

Output

```
(10, 20, 30, 40)
('Python', 'Java', 'C++')
(10, 'Python', 3.14, True)
```

Tuple with Single Element:-

```
t = (10,)
print(type(t))
```

Output

```
<class 'tuple'>
```

Note: without comma it consider as integer.

Accessing Tuple Elements (Indexing):-

```
t = (10, 20, 30, 40)
```

```
print(t[0])
print(t[2])
```

Output

```
10
30
```

Tuple Slicing:-

```
t = (10, 20, 30, 40, 50)
```

```
print(t[1:4])
```

Output

```
(20, 30, 40)
```

Python In-built functions for Tuple:-

Step	Syntax	Explanation
1	max()	To check maximum element in tuple
2	min()	To check minimum element in tuple
3	type()	To check datatype of object
4	id()	To check memory address of object
5	print()	To print output
6	len()	To count number of elements
7	tuple()	To create empty tuple or type casting
8	sum()	To calculate sum of tuple elements

Example:-

```
t = (10, 20, 30, 40)
```

```
print(max(t))
print(min(t))
print(len(t))
print(type(t))
print(sum(t))
```

Output

```
40
10
4
```

```
<class 'tuple'>
100
```

Tuple Methods

Tuple me methods bahut kam hote hain kyunki tuple immutable hota hai.

1. count()
2. index()

1. count():-- Count occurrences of element:-

```
t = (10, 20, 10, 30, 10)

print(t.count(10))

# O/P: 3
```

2. index():-- Find index of element:-

```
t = (10, 20, 30, 40)

print(t.index(30))

# O/P: 2
```

Tuple Packing:

```
t = 10, 20, 30

print(t)
print(type(t))

# Output
(10, 20, 30)
<class 'tuple'>
```

Tuple Unpacking:

```
t = (10, 20, 30)
a, b, c = t
print(a)
print(b)
print(c)

# Output
10
20
30
```

